Exercises learning task 15

2026 – 2027

Exercise 9.1: State whether each of the following situations is a correctly stated hypothesis testing problem and why.

(a) $H_0 : \mu = 25$, $H_1 : \mu \neq 25$
(b) $H_0 : \sigma > 10$, $H_1 : \sigma = 10$
(c) $H_0 : \bar{x} = 50$, $H_1 : \bar{x} \neq 50$
(d) $H_0 : p = 0.1$, $H_1 : p = 0.5$
(e) $H_0 : s = 20$, $H_1 : s < 20$

Exercise 9.5: A textile fiber manufacturer is investigating a new drapery yarn, which the company claims has a mean thread elongation of $12\,\mathrm{kg}$ with a standard deviation of $1.5\,\mathrm{kg}$. The company wishes to test the hypothesis $H_0 : \mu = 12$ against $H_1 : \mu < 12$, using a random sample of four specimens.

(a) What is the type I error probability if the critical region is defined as $\bar{x} < 11.5\,\mathrm{kg}$?

(b) Find β for the case in which the true mean elongation is $11.25\,\mathrm{kg}$.

(c) Find β for the case in which the true mean is $11.5\,\mathrm{kg}$.

Exercise 9.7: In Exercise 9.5, find the boundary of the critical region if the type I error probability is

(a) $\alpha = 0.01$ and $n = 5$
(b) $\alpha = 0.05$ and $n = 5$
(c) $\alpha = 0.01$ and $n = 20$
(d) $\alpha = 0.05$ and $n = 20$

Exercise 9.9: In Exercise 9.5, calculate the p-value if the observed statistic is

(a) $\bar{x} = 11.25$ (b) $\bar{x} = 11.0$ (c) $\bar{x} = 11.75$

Exercise 9.15: A consumer products company is formulating a new shampoo and is interested in foam height (in millimeters). Foam height is approximately normally distributed and has a standard deviation of $25\,\mathrm{mm}$. The company wishes to test $H_0 : \mu = 175\,\mathrm{mm}$ versus $H_1 : \mu > 175\,\mathrm{mm}$, using the results of $n = 15$ samples.

(a) Find the type I error probability α if the critical region is $\bar{x} > 185$.

(b) What is the probability of type II error if the true mean foam height is $185\,\mathrm{mm}$?

(c) Find β for the true mean of $195\,\mathrm{mm}$.

Exercise 9.17: In Exercise 9.15, find the boundary of the critical region if the type I error probability is

(a) $\alpha = 0.01$ and $n = 10$

Exercise 9.19: In Exercise 9.15, calculate the p-value if the observed statistic is

(a) $\bar{x} = 180$ (b) $\bar{x} = 190$ (c) $\bar{x} = 170$

Exercise 9.31: A hypothesis will be used to test that a population mean equals 4 against the alternative that the population mean is more than 4 with known standard deviation σ. What

is the critical value for the test statistic Z_0 for the following significance levels?

(a) 0.01 (b) 0.05 (c) 0.10

Exercise 9.33: For the hypothesis test $H_0 : \mu = 6$ against $H_1 : \mu \neq 6$ and variance known, calculate the p-value for each of the following test statistics.

(a) $z_0 = 2.05$ (b) $z_0 = -1.84$ (c) $z_0 = 0.4$